

Multi-Objective Mathematical Programming Software

Multiojective Optimization Problem (MOP) notations

$$\begin{array}{ll}\min & F(x) \\ \text{s.t.} & x \in X\end{array}$$

where:

$$\begin{array}{ll}X \subseteq \mathbb{R}^n & \longrightarrow \text{the set of feasible solutions} \\ Y = F(X) \subseteq \mathbb{R}^p & \longrightarrow \text{the set of images}\end{array}$$

Computing sets X_E and Y_N for MOP:

$$y = F(x)$$

$y^* \in Y$ is nondominated, if $\nexists y \in Y$ such that $y_i \leq y_i^*, \forall i$ and $y \neq y^*$.

Y_N is the set of nondominated points.

$x^* \in X$ is efficient if y^* is nondominated.

X_E is a complete set of efficient solutions.

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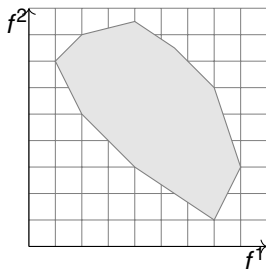
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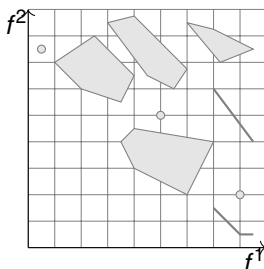
Examples of Y and Y_N when $p = 2$

Computing Y_N for Multiobjective (linear) Optimization Problems (MOP):

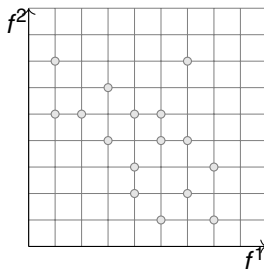
$$x \in \mathbb{R}^{n_1}$$



$$x \in \mathbb{R}^{n_1} \times \mathbb{Z}^{n_2}$$



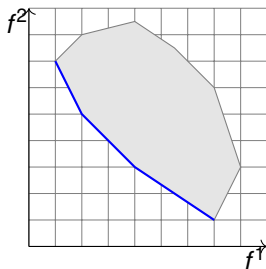
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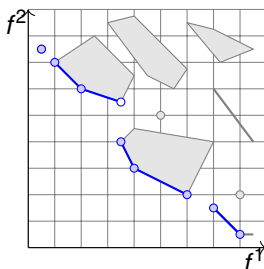
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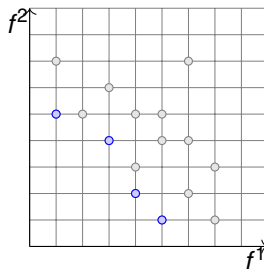
Continuous
nondominated set
(edges)

$$x \in \mathbb{R}^{n_1} \times \mathbb{Z}^{n_2}$$



Nondominated set
composed of edges
that are either closed,
half-open, open or
reduced to a point

$$x \in \mathbb{Z}^{n_2}$$

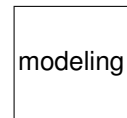


Discrete set of
nondominated
points

Multi-Objective Mathematical Programming Software

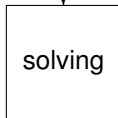
- JuMP
- GNU MP
- Pyomo...

- AMPL
- GAMS
- OPL...



low-level formulation
(flat form)

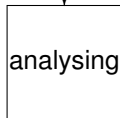
- MPS
- LP...



- HiGHS
- GLPK
- SCIP...

- GUROBI
- CPLEX
- Xpress...

high-level model
(structured form)



- ...

Software for modeling activity

Modelers: software tools devoted to express mathematical optimisation problems

- ▶ Algebraic Modeling Language (AML):

a high-level programming language designed for mathematical optimisation with an emphasis on declarative modeling and independent of any particular solver.

- ▶ expressed in proprietary languages:
GAMS, AMPL, AIMMS, OPL, Mosel, GMP, ZIMPL, ...
- ▶ close to programming languages:
JuMP (Julia), Pyomo (Python),...

- ▶ Solver-Oriented Interface (SOI):

tightly coupled to a specific optimisation solver; provide programmatic control and access to solver-specific features within a host programming language (C/C++, Java, Python,...)

- ▶ Application Programming Interfaces (API):
API C++ Gurobi,...
- ▶ embedded Domain-Specific Languages (DSL) :
highspy, MATLAB, Hexaly Modeler,...

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Software for solving activity

Optimisation solver: a specialized software engine designed to find solutions to mathematical programming problems

- ▶ Generic MILP solvers:

implement numerical algorithms tailored to problem classes (LP, MIP,...)

- ▶ Gurobi, CPLEX, Xpress, MOSEK, Hexaly, COPT,...
- ▶ GLPK, SCIP, HiGHS, CBC/CLP,...

- ▶ Specific MOMP solvers:

specialized optimisation engines designed to directly handle problems with multiple objectives

- ▶ ADBASE, BENSOLVE, INNER, PaMILO, PolySCIP, vOptSpecific

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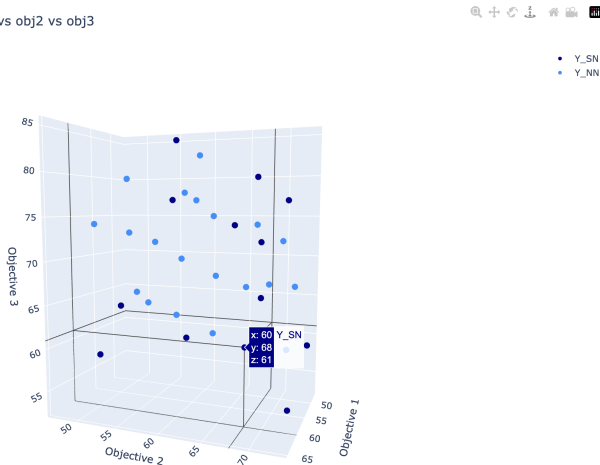
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Software for analysing activity

Homemade...

Objective space : obj1 vs obj2 vs obj3



Summary of software examined



<i>Software</i>	<i>Commercial</i>	<i>Open-source</i>	<i>Modeler</i>	<i>Solver</i>	<i># Trade-offs</i>
MATLAB	✓	–	SOI	–	≥1
Gurobi Optimizer	✓	–	SOI	generic	1
HiGHS	–	✓	SOI	generic	1
COPT	✓	–	SOI	generic	1
Hexaly	✓	–	SOI	generic	1
FICO Xpress Optimizer	✓	–	AML+SOI	generic	1
CPLEX Optimisation Studio	✓	–	AML+SOI	generic	1
AMPL	✓	–	AML	–	1
AIMMS	✓	–	AML	–	1
GAMS	✓	–	AML	–	≥1
Pyomo	–	✓	AML	–	–
PuLP	–	✓	AML	–	–
CVXPY	–	✓	AML	specific	1
MOCVPY	–	✓	AML	specific	≥1
BENSOLVE	–	✓	–	specific	≥1
INNER	–	✓	–	specific	≥1
PaMILO	–	✓	–	specific	≥1
SCIP with ZIMLP and PolySCIP	–	✓	AML	specific	≥1
JuMP with MultiObjectiveAlgorithms	–	✓	AML	meta-solver	≥1